Bhautik Lukhi

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Bielefeld, Germany

Education

M.Sc. in Data Science — Bielefeld, Germany
Bielefeld University — *Apr 2023 – Present*

B.E. in Computer Engineering — Bhavnagar, India
Bhavnagar University — *July 2018 – June 2022*

Experience

Research Assistant — May 2024 – Feb 2026
Bielefeld University, Dept. of Sports Science — *Bielefeld, Germany*

- Processed and analyzed total **15M+ rows** of structured and semi-structured sports event data using Python and web scraping tools.
- Contributed to research design, literature reviews, and experimental planning, actively supporting academic publication efforts.
- Designed and applied statistical and ML models to uncover patterns in referee decision-making and bias.
- Engineered features from **90k+ NBA L2M reports** to investigate the **make-up calls** and officiating biases.

Research Assistant — May 2024 – Aug 2024
Hochschule Bielefeld, CareTech OWL Project — *Bielefeld, Germany*

- Developed privacy-preserving algorithms for real-world sensor data using **Fully Homomorphic Encryption (FHE)** and **Concrete-ML**.

Teaching Assistant - Mathematics — Oct 2023 – Mar 2024
Bielefeld University — *Bielefeld, Germany*

- Conducted **3 weekly tutorials** for undergraduate students.
- Contributed to the highest-ever passing rate for the course as part of a **team of 5 tutors**.

Skills

Programming: Python, R, Java/JavaScript, LaTeX
ML/AI: Scikit-learn, TensorFlow, PyTorch, Transformers, LLMs, Generative AI, OpenCV
Data Tools: NumPy, pandas, Plotly, Seaborn, Matplotlib, Beautiful Soup
Deployment: Streamlit, Docker, Git, GitHub
Soft Skills: Teamwork, Communication, Troubleshooting, Adaptability

Projects

Squat Form Guru — Python, OpenCV, Mediapipe, Node.js [see here]

- Built a computer vision tool to analyze squat form using **Mediapipe** pose estimation.
- Deployed as a web app to detect form errors and provide automatic feedback.

Bundesliga 2v1 Tactical Analysis — Python, Pandas, Scikit-learn, Matplotlib [see here]

- Analyzed Bundesliga data to identify patterns in 2vs1 situations for **FC Schalke 04.**
- Performed feature engineering, pattern recognition, clustering for visualized tactical behaviors.

Salary Prediction Web App — Scikit-learn, Streamlit, Pandas [see here]

- Developed a regression-based ML model to predict developer salaries using Stack Overflow data.
- Performed feature engineering and model tuning using **GridSearchCV**.

Languages

English, Gujarati, Hindi — **Fluent**

German — **Intermediate (B1)**